# 10

# Diagonalisation

**Historical Note**

Once eigenvalues were understood, the obvious question was whether a change of basis could always reduce a matrix to its simplest possible form, and the answer turned out to be delicate enough to occupy the best algebraists of the nineteenth century. Cauchy had settled the symmetric case in 1829, where the answer is always yes. The general case was resolved twice, almost at once and by quite different routes: Weierstrass in 1868 with his theory of elementary divisors, and Camille Jordan in his *Traité des substitutions* of 1870 with the canonical form that carries his name. The two accounts are equivalent, the priority was disputed at the time, and posterity has been inconsistent, attaching Jordan's name to the form and Weierstrass's to the invariants that determine it. What both showed is that diagonal is not always reachable, that the obstruction is precisely the gap in Theorem 9.16, and that there is a next-best form which always is. This chapter treats the case where diagonal works, and Chapter 11 treats the next-best form.

Chapter 9 found the directions along which an operator merely stretches. If there are enough of them to form a basis, then in that basis the operator does nothing but scale each coordinate independently, and every question about it becomes arithmetic. This chapter says when that happens, what it is worth, and what to do when it does not.

## 10.1 When is a matrix diagonal?

**Definition 10.1** (Diagonalisable)**.** *An operator $T$ on a finite-dimensional space $V$ is* diagonalisable *if there is a basis $B$ of $V$ for which $[T]_B$ is diagonal. A matrix is diagonalisable if it is similar to a diagonal matrix, that is, if $A = PDP^{-1}$ for some invertible $P$ and diagonal $D$.*

**Proposition 10.2** (Diagonal means eigenbasis). *Let $B = (v_1, \dots, v_n)$ be an ordered basis of $V$. Then $[T]_B$ is diagonal if and only if every $v_j$ is an eigenvector of $T$, and in that case the diagonal entries are the corresponding eigenvalues.*

***Proof.*** By Definition 7.17, the $j$th column of $[T]_B$ is $[Tv_j]_B$. That column has its only nonzero entry in position $j$, say $d_j$, precisely when $Tv_j = d_j v_j$, which for $v_j \neq \mathbf{0}$ is precisely the statement that $v_j$ is an eigenvector with eigenvalue $d_j$. □

**Theorem 10.3** (The factorisation). *A matrix $A \in M_n(\mathbb{R})$ is diagonalisable if and only if $\mathbb{R}^n$ has a basis of eigenvectors of $A$. In that case, if $P$ has those eigenvectors as its columns and $D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ lists the matching eigenvalues in the same order, then*

$$A = PDP^{-1}, \qquad \text{equivalently} \qquad P^{-1}AP = D.$$

***Proof.*** Write $v_1, \dots, v_n$ for the columns of $P$. By Proposition 4.4 the $j$th column of $AP$ is $Av_j$, while the $j$th column of $PD$ is $\lambda_j v_j$. So

$$AP = PD \iff Av_j = \lambda_j v_j \text{ for every } j.$$

Moreover $P$ is invertible exactly when its columns are independent, by Theorem 4.16, that is, exactly when they form a basis. Combining the two statements gives the theorem in both directions: from an eigenbasis we build $P$ and get $A = PDP^{-1}$, and from $A = PDP^{-1}$ we read off that the columns of $P$ are $n$ independent eigenvectors. □

**Key Idea**

The three matrices in $A = PDP^{-1}$ have three different jobs, and reading them right to left tells you what $A$ does: $P^{-1}$ rewrites a vector in the eigenbasis, $D$ scales each of those coordinates by its own eigenvalue, and $P$ translates the answer back into the standard basis. Diagonalising is not a computation performed on $A$; it is a change of viewpoint from which $A$ was always this simple.

Figure 10.1 draws those three jobs as a square of arrows: going across the top is what $A$ does, and going down, across the bottom and back up is what $PDP^{-1}$ does.

**Example 10.4** (A diagonalisation, in full). *Take $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ from Example 9.7, with eigenvalues 3 and 1 and eigenvectors $(1, 1)$ and $(1, -1)$. Put*

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad D = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

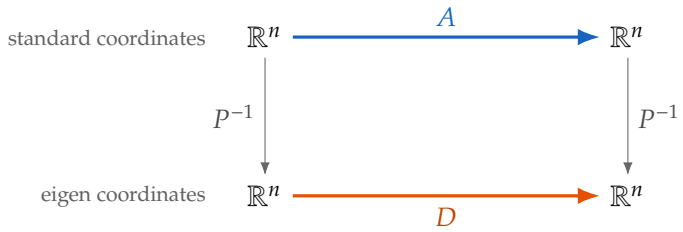



**Figure 10.1.** Theorem 10.3 as a diagram. The same space appears twice on each side, described in two different coordinate systems, and $P^{-1}$ is the translation between them. Going across the top is what $A$ does; going down, across the bottom and back up is what $PDP^{-1}$ does, and the content of the theorem is that these are the same journey.

*Since $P^2 = 2\mathbf{1}$, we have $P^{-1} = \frac{1}{2}P$, which makes the check easy:*

$$PD = \begin{pmatrix} 3 & 1 \\ 3 & -1 \end{pmatrix}, \qquad PDP^{-1} = \tfrac{1}{2}\begin{pmatrix} 3 & 1 \\ 3 & -1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \tfrac{1}{2}\begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix} = A.$$

*Note that $P$ is not unique: reordering the columns reorders $D$ to match, and scaling any column changes nothing at all, since an eigenvector stays an eigenvector when scaled.*

## 10.2 When can it be done?

**Corollary 10.5** (The easy sufficient condition)**.** *If $A \in M_n(\mathbb{R})$ has $n$ distinct eigenvalues then it is diagonalisable.*

***Proof.*** Pick one eigenvector for each. They are independent by Theorem 9.13, hence a basis by Theorem 6.14(c), and Theorem 10.3 applies. □

**Caution**

The condition is sufficient and not necessary, and confusing the two is a standard error. The identity matrix has the single eigenvalue 1 and is already diagonal; the matrix of Example 9.8 has a repeated eigenvalue and is diagonalisable. What matters is not whether eigenvalues repeat but whether the eigenspaces are large enough, which is what the next theorem measures.

**Lemma 10.6** (Independence across eigenspaces)**.** *Let $\lambda_1, \dots, \lambda_k$ be distinct eigenvalues of $T$ and, for each $i$, let $S_i$ be an independent subset of $E_{\lambda_i}$. Then $S_1 \cup \cdots \cup S_k$ is independent.*

***Proof.*** Suppose a vanishing linear combination of the whole union is given, and let $w_i$ be the part of it coming from $S_i$, so that $w_i \in E_{\lambda_i}$ and $w_1 + \cdots + w_k = \mathbf{0}$. If some $w_i$ were nonzero, then discarding the zero ones would leave a vanishing sum of eigenvectors for distinct eigenvalues, which Theorem 9.13 forbids. So every $w_i = \mathbf{0}$, and then the independence of each $S_i$ forces all its coefficients to vanish. □

**Theorem 10.7** (The criterion)**.** *Let $T$ be an operator on an $n$-dimensional space whose characteristic polynomial splits into linear factors, which over $\mathbb{C}$ is automatic. Write $\lambda_1, \dots, \lambda_k$ for the distinct eigenvalues, $g_i = \dim E_{\lambda_i}$ and $m_i$ for the algebraic multiplicities. The following are equivalent.*

*(a) $T$ is diagonalisable.*

*(b) $g_1 + \cdots + g_k = n$.*

*(c) $g_i = m_i$ for every $i$.*

***Proof.*** (a) $\Rightarrow$ (b). Let $B$ be an eigenbasis, so $|B| = n$ and every member of $B$ lies in some $E_{\lambda_i}$. Those members of $B$ lying in $E_{\lambda_i}$ are independent vectors of $E_{\lambda_i}$, so there are at most $g_i$ of them; summing over $i$ accounts for all of $B$ and gives $n \le \sum_i g_i$. In the other direction, a union of bases of the eigenspaces is independent by Lemma 10.6, and an independent set has at most $n$ elements by Theorem 6.14(a), so $\sum_i g_i \le n$. Hence equality.

(b) $\Rightarrow$ (a). Take a basis of each eigenspace and form the union. It is independent by Lemma 10.6, and it has $\sum_i g_i = n$ elements, so it is a basis, by Theorem 6.14(c). Every one of its members is an eigenvector, so Proposition 10.2 applies.

(b) $\iff$ (c). Because the characteristic polynomial splits, $\sum_i m_i = n$. By Theorem 9.16 we have $g_i \le m_i$ for every $i$, so $\sum_i g_i \le \sum_i m_i = n$ with equality if and only if $g_i = m_i$ term by term. □

**Example 10.8** (Diagonalisable with a repeat)**.** *For $A = \left(\begin{smallmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{smallmatrix}\right)$ of Example 9.8 the eigenvalues are 5 and 2, with $m_5 = 1$, $m_2 = 2$ and $g_5 = 1$, $g_2 = 2$. Since $g_i = m_i$ throughout, $A$ is diagonalisable. Taking $(1, 1, 1)$ for $E_5$ and the independent pair $(1, -1, 0)$,*

$(1, 0, -1)$ *for* $E_2$,

$$P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix}, \qquad D = \operatorname{diag}(5, 2, 2), \qquad A = PDP^{-1}.$$

**Example 10.9** (Defective matrices)**.** *The shear* $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ *has* $m_1 = 2$ *but* $g_1 = 1$ *by Example 9.17, so it is not diagonalisable. Neither is*

$$N = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix},$$

*whose only eigenvalue is* 2 *with* $m_2 = 3$*, while* $N - 2\mathbf{1} = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{smallmatrix}\right)$ *has rank* 2 *and hence* $g_2 = 3 - 2 = 1$*. A matrix with* $g_i < m_i$ *for some* $i$ *is called* defective*. There is nothing pathological about them; they are simply the matrices for which the next-best form is needed.*

**Key Idea**

Every real symmetric matrix is diagonalisable, its eigenvalues are real, and the eigenbasis can be chosen orthonormal. That is the spectral theorem, and it is the reason both worked examples of this chapter came out so cleanly: $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ and the $3 \times 3$ above are symmetric, so they never had a chance of being defective. We prove it in Part V, once inner products are available, because the orthogonality is the heart of the statement and cannot be seen without them.

## 10.3 What diagonalisation is for

**Proposition 10.10** (Powers and polynomials)**.** *If* $A = PDP^{-1}$ *then* $A^k = PD^kP^{-1}$ *for every* $k \geq 0$*, and more generally* $q(A) = P\,q(D)\,P^{-1}$ *for every polynomial* $q$*. Since* $D$ *is diagonal,* $D^k$ *and* $q(D)$ *are computed entry by entry along the diagonal.*

***Proof.*** Write $A^k$ as the product of $k$ copies of $PDP^{-1}$. Each adjacent pair $P^{-1}P$ cancels, leaving $PD^kP^{-1}$. Taking linear combinations of these gives the statement for an arbitrary polynomial, and a diagonal matrix is raised to a power entry by entry. □

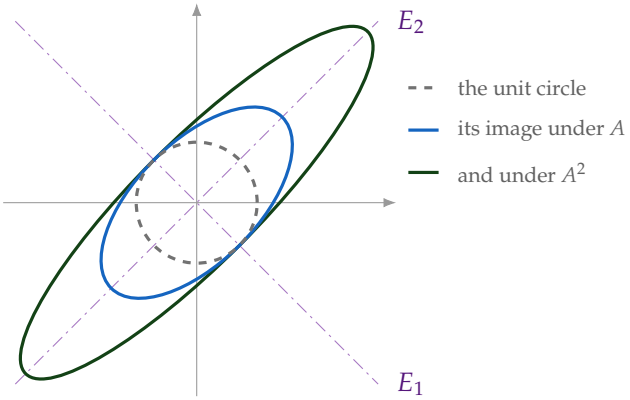



**Figure 10.2.** Powers of a diagonalisable matrix, here $A = \left(\begin{smallmatrix} 1.5 & 0.5 \\ 0.5 & 1.5 \end{smallmatrix}\right)$ with eigenvalues 2 along $E_2$ and 1 along $E_1$. Each application multiplies one axis by 2 and leaves the other alone, so the circle becomes an ever thinner ellipse pointing along the dominant eigendirection. This is exactly why the power method of Chapter 9 works, and why it converges at the rate $|\lambda_2/\lambda_1|^k$.

**Example 10.11** (Fibonacci in closed form)**.** *Let $F_0 = 0$, $F_1 = 1$ and $F_{n+1} = F_n + F_{n-1}$. The recurrence is the matrix statement*

$$\begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = A \begin{pmatrix} F_n \\ F_{n-1} \end{pmatrix}, \qquad A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix},$$

*so $\left(\begin{smallmatrix} F_{n+1} \\ F_n \end{smallmatrix}\right) = A^n \left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right)$. Now $p_A(\lambda) = \lambda^2 - \lambda - 1$, whose roots are the golden ratio $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$. They are distinct, so $A$ is diagonalisable by Corollary 10.5, and $A^n = PD^nP^{-1}$ with $D^n = \operatorname{diag}(\varphi^n, \psi^n)$. Every entry of $A^n$ is therefore a fixed combination of $\varphi^n$ and $\psi^n$, and matching the two initial values gives Binet's formula*

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}.$$

*Check it at $n = 1$: $\varphi - \psi = \sqrt{5}$, so $F_1 = 1$. At $n = 2$: $\varphi^2 - \psi^2 = (\varphi + \psi)(\varphi - \psi) = 1 \cdot \sqrt{5}$, so $F_2 = 1$. Since $|\psi| < 1$, the second term vanishes rapidly and $F_n$ is the nearest integer to $\varphi^n/\sqrt{5}$. A recursion has become a formula, and the eigenvalues are the reason.*

**Example 10.12** (Linear differential equations)**.** *Define $e^A = \sum_{k \geq 0} A^k/k!$, a series that converges entry by entry for every square $A$. If $A = PDP^{-1}$ then Proposition 10.10*

*applies term by term and*

$$e^{At} = P\,e^{Dt}\,P^{-1} = P\,\operatorname{diag}\bigl(e^{\lambda_1 t},\dots,e^{\lambda_n t}\bigr)P^{-1}.$$

*The system $x'(t) = Ax(t)$ with $x(0) = x_0$ then has solution $x(t) = e^{At}x_0$, which in the eigenbasis reads*

$$x(t) = c_1 e^{\lambda_1 t}v_1 + \cdots + c_n e^{\lambda_n t}v_n,$$

*where the $c_i$ are the coordinates of $x_0$ in that basis. Each eigendirection evolves entirely on its own, and the sign of $\lambda_i$ decides whether that component grows or decays. This is the reason Example 5.3(vi) had a two-dimensional solution space, and it is why stability questions in dynamics reduce to asking where the eigenvalues sit.*

**Insight**

A Markov chain is a matrix of transition probabilities whose columns sum to 1. Such a matrix always has 1 as an eigenvalue, with the stationary distribution as its eigenvector, and if every other eigenvalue is smaller in absolute value then $A^k$ converges to the projection onto that one direction. Figure 10.2 is the picture of the convergence, and Proposition 10.10 is the proof: in the eigenbasis, everything except the dominant component is multiplied by $\lambda_i^k \to 0$. The rate is governed by the ratio $|\lambda_2/\lambda_1|$, called the spectral gap, and the entire practical difference between a chain that mixes in ten steps and one that mixes in ten million is the size of that gap.

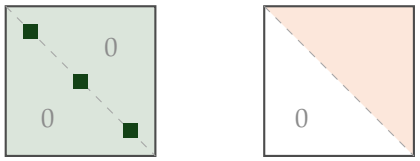
## 10.4 When diagonalisation is the wrong tool

Figure 10.3 sets the two normal forms against each other. Diagonal is what one would like and cannot always have; upper triangular is what one can always have over $\mathbb{C}$, and by a unitary change of basis at that.

**Caution**

Diagonalisation is a poor foundation for computation, for a reason more subtle than the cost. The factorisation $A = PDP^{-1}$ is only as good as $P$, and $P$ can be arbitrarily badly conditioned. Consider

$$A_\varepsilon = \begin{pmatrix} 1 & 1 \\ 0 & 1+\varepsilon \end{pmatrix}.$$




$D$
diagonal
only sometimes

$T$
triangular
always, over $\mathbb{C}$

**Figure 10.3.** Two normal forms. Diagonal is what we would like and cannot always have; upper triangular is what we can always have over $\mathbb{C}$, by Schur's theorem, and with the change of basis given by a unitary $Q$, which costs nothing in accuracy. Numerical libraries compute the second and only compute the first when the matrix is symmetric, where the two coincide.

For every $\varepsilon \neq 0$ this has distinct eigenvalues and is diagonalisable, but its eigenvectors are $(1, 0)$ and $(1, \varepsilon)$, which become parallel as $\varepsilon \to 0$; the condition number of $P$ grows like $1/\varepsilon$. At $\varepsilon = 0$ the matrix is the defective shear and no $P$ exists at all. So diagonalisability is not a stable property: an arbitrarily small perturbation of a defective matrix is diagonalisable, and an arbitrarily small perturbation of a diagonalisable matrix can be defective. Any algorithm whose output is $P$ inherits that instability.

**Key Idea**

What numerical practice uses instead is Schur's theorem: over $\mathbb{C}$ every square matrix satisfies $A = QTQ^*$ with $T$ upper triangular and $Q$ unitary, meaning $Q^*Q = \mathbf{1}$. Triangular is almost as good as diagonal, since the eigenvalues still sit on the diagonal by Proposition 9.18(c); and unitary changes of basis preserve lengths, so they cannot amplify error. That is the form the QR algorithm of Section 9.6 actually converges to. We prove it in Part V, where the word unitary can be defined.

**Open Problem 10.13** (The variety of commuting triples)**.** *Let $C_k(n)$ be the set of $k$-tuples of $n \times n$ complex matrices that commute pairwise. It is an algebraic variety, cut out by the polynomial equations $A_iA_j = A_jA_i$.* Is $C_3(n)$ irreducible, that is, not a union of two smaller varieties?
What is known. *For $k = 2$ the answer is yes, a theorem of Motzkin and Taussky, and the proof is essentially the observation that a generic commuting pair is simultaneously*

*diagonalisable and that the simultaneously diagonalisable pairs form a dense irreducible set. For $k \geq 4$ the answer is no, and explicit reducible examples are known for $n \geq 4$.* The case $k = 3$ is open*: it is known to be irreducible for $n \leq 10$ or so, by increasingly heavy computation, and unknown beyond [285, 330].*

Why it is a question about this chapter. *Diagonalisation says a single generic matrix is determined up to similarity by its eigenvalues, and simultaneous diagonalisation says the same for a commuting family,* provided *each member is diagonalisable. Non-diagonalisable commuting tuples are the entire content of the problem: they are the points where the dense simultaneously diagonalisable stratum might fail to be dense. The equivalent and more suggestive formulation is whether every triple of commuting matrices can be approximated by a triple of commuting diagonalisable ones.*

Where a reader might start. *Prove the $k = 2$ case for $n = 2$ by hand: show that any two commuting $2 \times 2$ matrices are a limit of simultaneously diagonalisable pairs. Then take three commuting nilpotent $4 \times 4$ matrices, for instance those spanned by products of two commuting Jordan blocks, and try the same. Chapter 11 explains what the obstruction looks like, and Chapter 98 is the language in which these varieties are studied. Figure 10.4 shows the phenomenon in the smallest case that has anything to show: two commuting symmetric matrices, diagonalised by one and the same pair of directions.*

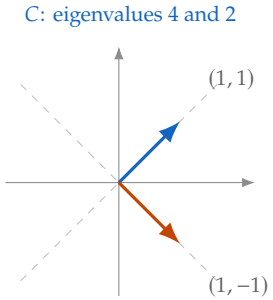



**Figure 10.4.** Commuting matrices share an eigenbasis. Here $C = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right)$ and $D = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$ satisfy $CD = DC = \left(\begin{smallmatrix} 5 & 7 \\ 7 & 5 \end{smallmatrix}\right)$, and both are diagonalised by the same two directions: $C$ acts as 4 and 2 along them, $D$ as 3 and $-1$. Commuting is not a technical hypothesis but exactly the condition under which one change of coordinates serves both, and the proof is short: if $Cv = \lambda v$ then $C(Dv) = D(Cv) = \lambda(Dv)$, so $D$ preserves each eigenspace of $C$ and can be diagonalised inside it. Take $\mathrm{diag}(2, 3)$ and the swap matrix instead, whose products are $\left(\begin{smallmatrix} 0 & 2 \\ 3 & 0 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 0 & 3 \\ 2 & 0 \end{smallmatrix}\right)$, and no such shared basis exists.

## Exercises

**Exercise 10.14.** *For each matrix, decide whether it is diagonalisable, and if so exhibit $P$ and $D$ with $A = PDP^{-1}$.*

*(a)* $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$

*(b)* $\begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$

*(c)* $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$

*(d)* $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

**Exercise 10.15.** *Using your answer to Exercise 10.14(a), compute $A^{10}$ without multiplying ten matrices together. Then find $\lim_{k\to\infty} (A/3)^k$ and identify the limit geometrically.*

**Exercise 10.16.** *Give a $3 \times 3$ matrix that is diagonalisable and has only two distinct eigenvalues, and one that has only two distinct eigenvalues and is not diagonalisable. What is the difference between them, in the language of Theorem 10.7?*

**Exercise 10.17.** *Let $A$ be diagonalisable with all eigenvalues positive. Show that there is a matrix $S$ with $S^2 = A$, and that if all eigenvalues are distinct there are exactly $2^n$ such $S$ built this way.* Hint: what is a square root of a diagonal matrix?

**Exercise 10.18.** *Suppose $A$ and $B$ are both diagonalisable and $AB = BA$. Show that they can be diagonalised by the* same *$P$.* Hint: by Exercise 9.26, $B$ preserves each eigenspace of $A$; now diagonalise $B$ inside each one.

**Exercise 10.19.** *Let $G_0 = 2$, $G_1 = 1$ and $G_{n+1} = G_n + G_{n-1}$, the Lucas numbers. Using the same matrix as Example 10.11, show that $G_n = \varphi^n + \psi^n$, and verify it for $n = 0, 1, 2, 3$.*

**Exercise 10.20.** *Solve $x' = Ax$ with $A = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$ and $x(0) = (1, 0)$, using Example 10.12. Describe the behaviour as $t \to \infty$, and say which eigenvalue is responsible.*

**Exercise 10.21.** *Let $A$ be diagonalisable with eigenvalues $\lambda_1, \dots, \lambda_n$. Prove that* $\operatorname{trace}(A^k) = \sum_i \lambda_i^k$ *for every $k \geq 1$. Then explain why this is still true when $A$ is defective.* Hint for the second part: use the triangular form of the Key Idea in Section 10.4.

**Exercise 10.22.** *Take $A_\varepsilon = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1+\varepsilon \end{smallmatrix}\right)$ from the Caution above. Compute $P$ and $P^{-1}$ explicitly as functions of $\varepsilon$, and confirm that the entries of $P^{-1}$ blow up like $1/\varepsilon$. What does that say about computing $A_\varepsilon^{100}$ through the diagonalisation when $\varepsilon = 10^{-8}$?*

## References and further reading

The Weierstrass and Jordan story is a good illustration of how a subject settles: two complete solutions in three years, by different methods, followed by a century of textbooks attributing the result inconsistently. Axler [37] organises the whole of this material around invariant subspaces rather than around $P$ and $D$, which makes the passage to the Jordan form considerably smoother than the matrix-first route we have taken, and is worth reading for that reason. Hefferon [307] has a large supply of diagonalisation exercises with the arithmetic arranged to come out cleanly. Banica [51], our baseline, treats diagonalisation as a stepping stone to the spectral theorems of Part V and is brisk about it. For the numerical picture, and in particular for why the eigenvector matrix is the wrong object to compute and the Schur form is the right one, Trefethen and Bau [607] is again the reference; their treatment of the sensitivity of eigenvalues to perturbation is the natural sequel to the Caution in Section 10.4.